# The Math and Physics of MechanicsDSL

*A complete account of what this library computes,*
*and why it is built the way it is.*

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## How to read this document

No prior calculus is assumed. Section 1 establishes the notation, and unfamiliar constructions are explained where they first appear. Readers already familiar with Lagrangian mechanics may proceed directly to Section 5.

The document divides in two:

- **Sections 1–7** describe the core engine and are intended to be read in order.
- **Section 8** is a reference catalog of the sixteen physics domains the library implements. Each entry opens with a one-sentence statement of scope; the material beneath documents coverage rather than inviting continuous reading.

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## Contents

# 1 Notation

**Symbols denote quantities.** Here $m$ is mass, $l$ length, $g$ gravitational acceleration ($9.81\ \mathrm{m/s^2}$ at Earth's surface), $t$ time, and $E$ energy. Greek letters follow the same convention: $\theta$ typically an angle, $\omega$ an angular frequency.

$q$ **denotes a generalized coordinate** — a single number specifying the configuration of a system. For a pendulum this might be the angle from vertical; for a block on a track, the distance along it. Where several are required they are indexed $q_1, q_2, \ldots$. The term *generalized* signals that no particular kind of coordinate is assumed: an angle and a distance are treated identically.

**An overdot denotes differentiation with respect to time** — the rate at which a quantity changes, per second.

| Symbol | Read as | Quantity |
|---|---|---|
| $q$ | "$q$" | Position |
| $\dot{q}$ | "$q$-dot" | Velocity — the rate of change of position |
| $\ddot{q}$ | "$q$-double-dot" | Acceleration — the rate of change of velocity |

For a pendulum with angular coordinate $\theta$, then, $\dot{\theta}$ is its angular velocity and $\ddot{\theta}$ its angular acceleration. This is the entire convention.

**The partial derivative** $\partial f/\partial x$ measures how a quantity $f$ responds to variation in $x$ while every other variable is held fixed. On a topographic map, $\partial(\text{elevation})/\partial(\text{easting})$ is the gradient one would measure walking due east; variation in the north–south direction contributes nothing. The notation isolates a single direction of dependence.

**The operator** $\mathrm{d}/\mathrm{d}t$ denotes the total rate of change with respect to time, accounting for every quantity that varies as time passes.

**Additional notation:**

| Symbol | Meaning |
|---|---|
| $\sum$ | Sum over the indicated terms |
| $\int$ | Integral — a continuous sum |
| $\nabla$ | Gradient — direction and magnitude of steepest increase |
| $\approx$ | Approximately equal |
| $\delta_{ij}$ | Equal to 1 when $i = j$, otherwise 0 |

Nothing beyond this is assumed.

## 2 What the library does

A physical system is specified by a single expression — its **Lagrangian**. From that expression the library derives the equations of motion, integrates them numerically, and can translate them into thirteen target languages.

A complete pendulum specification:

```
\system{pendulum}
\defvar{theta}{Angle}{rad}
\parameter{m}{1.0}{kg}
\parameter{l}{1.0}{m}
\parameter{g}{9.81}{m/s^2}
\lagrangian{\frac{1}{2}*m*l^2*\dot{theta}^2 - m*g*l*(1 - \cos{theta})}
\initial{theta=0.5, theta_dot=0.0}
```

The declarations name the system, identify `theta` as the generalized coordinate, fix the physical parameters, state the Lagrangian, and set initial conditions. Everything downstream — equations of motion, energy expressions, generated C++ — follows from the Lagrangian line alone. No per-system code is written by hand.

The pipeline comprises five stages:

| Stage | Function | Module |
|---|---|---|
| Parse | Source text to abstract syntax tree | `parser/core.py` |
| Symbolize | Syntax tree to symbolic algebra | `symbolic.py` |
| Derive | Lagrangian to equations of motion | `symbolic.py` |
| Solve | Equations to numerical trajectory | `solver/` |
| Emit | Equations to source in a target language | `codegen/` |

# 3 Energy formulations versus force formulations

## 3.1 The two approaches

Classical mechanics admits two equivalent formulations. They yield identical predictions and differ in what they require of the person applying them.

**The Newtonian formulation** works with forces:

$$\vec{F} = m\vec{a} \tag{1}$$

Applying it requires identifying every force acting on every body and resolving each into components along chosen axes. The bookkeeping is vectorial, and it is the user's responsibility: an omitted force yields a wrong answer with no indication that anything is missing.

**The Lagrangian formulation** works with energy. Define a single scalar quantity, the Lagrangian:

$$L = T - V \tag{2}$$

where $T$ is the kinetic energy — the energy associated with motion — and $V$ is the potential energy, the energy associated with configuration. A raised mass and a compressed spring both store potential energy; both convert it to kinetic energy when released.

$L$ is a scalar. It has no direction and no components.

## 3.2 Why the difference rather than the sum

The subtraction is not a convention chosen for convenience; it is fixed by the variational principle underlying the formulation.

Among all conceivable paths a system might follow between two configurations, the physical path is distinguished by a property of the **action** — the integral of $L$ along the path. The physical path is the one for which the action is **stationary**: perturb the path infinitesimally in any direction and the action is unchanged to first order.

The analogy is to a minimum of a function. A ball at the base of a bowl sits where displacement in any direction produces no first-order change in height. The physical trajectory occupies the analogous position within the space of all possible trajectories.

Taking $T - V$ is what makes the stationary-action condition reproduce Newtonian dynamics. The result is the Euler–Lagrange equation, derived next.

Using the library requires no commitment to this justification. It does explain the minus sign.

## 3.3 The Euler–Lagrange equation

For each generalized coordinate $q$, the equation of motion is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0 \tag{3}$$

Read from the inside outward, this prescribes three operations:

1. $\partial L/\partial \dot{q}$ — differentiate $L$ with respect to the velocity.
2. $\mathrm{d}/\mathrm{d}t\,(\cdot)$ — differentiate that result with respect to time.
3. $\partial L/\partial q$ — differentiate $L$ with respect to the coordinate itself, and subtract.

The procedure is purely mechanical and identical for every system and every coordinate. That invariance is what makes it automatable, and automating it is what this library is.

### 3.4 Worked example: the simple pendulum

Applying the procedure once, in full, to the system specified in Section 2.

**Setup.** A point mass $m$ on a rigid massless rod of length $l$, displaced by angle $\theta$ from vertical.

The mass travels on a circular arc of radius $l$, so its speed is $l\dot{\theta}$. Kinetic energy is $\frac{1}{2}mv^2$:

$$T = \tfrac{1}{2}ml^2\dot{\theta}^2 \tag{4}$$

Displacement through angle $\theta$ raises the mass by $l(1-\cos\theta)$, giving gravitational potential energy $mgh$:

$$V = mgl\,(1-\cos\theta) \tag{5}$$

The Lagrangian is their difference:

$$L = \tfrac{1}{2}ml^2\dot{\theta}^2 - mgl(1-\cos\theta) \tag{6}$$

**Step 1: differentiate with respect to $\dot{\theta}$.** Only the kinetic term depends on $\dot{\theta}$. Differentiating a squared quantity $x^2$ yields $2x$, so $\frac{1}{2}ml^2\dot{\theta}^2$ yields $\frac{1}{2}ml^2 \cdot 2\dot{\theta}$:

$$\frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta} \tag{7}$$

This quantity is the pendulum's **angular momentum**. It emerges from the procedure without having been sought.

**Step 2: differentiate with respect to time.** Here $m$ and $l$ are constants, so only $\dot{\theta}$ varies; its time derivative is $\ddot{\theta}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ml^2\dot{\theta}\right) = ml^2\ddot{\theta} \tag{8}$$

**Step 3: differentiate with respect to $\theta$ and subtract.** Only the potential term depends on $\theta$. Expanding, $-mgl(1-\cos\theta) = -mgl + mgl\cos\theta$; the constant contributes nothing, and the derivative of $\cos\theta$ is $-\sin\theta$:

$$\frac{\partial L}{\partial \theta} = -mgl\sin\theta \tag{9}$$

Assembling:

$$ml^2\ddot{\theta} + mgl\sin\theta = 0 \tag{10}$$

Dividing by $ml^2$:

$$\boxed{\ddot{\theta} = -\frac{g}{l}\sin\theta} \tag{11}$$

**Interpretation.** Angular acceleration is proportional to $-\sin\theta$. The negative sign gives a restoring effect — acceleration is always directed back toward vertical — and the magnitude grows with displacement. Note that $m$ has cancelled: the period is independent of the mass, as observed.

**Verification.** For small displacements $\sin\theta \approx \theta$, reducing the equation to $\ddot{\theta} = -(g/l)\theta$, the equation of simple harmonic motion with angular frequency $\omega = \sqrt{g/l}$ and period

$$T = 2\pi\sqrt{l/g} \tag{12}$$

which is the standard result. It follows from the definition of $T - V$ and three differentiations.

*No force was identified at any point, and the rod's tension never appeared.* That observation motivates the next section.

# 4 Why the Lagrangian formulation

Both formulations are correct. The question is which can be carried out by software on the user's behalf. Six considerations follow, then the cases where the choice is a liability.

## 4.1 A scalar expression is representable as text

$L = T - V$ is a single algebraic expression. It can be tokenized, parsed, differentiated symbolically, and simplified by a computer algebra system.

The Newtonian formulation requires that the analysis already be done: forces enumerated, components resolved, frame selected. There is no natural textual representation of "the tension in the rod" from which software could infer the intended physics. There is a natural textual representation of kinetic energy.

## 4.2 Constraint forces do not appear

The rod in Section 3.4 exerts a tension on the mass. A Newtonian treatment must account for it — despite its doing no work, having no influence on the period, and being irrelevant to the question asked. It would be solved for and then eliminated.

In Section 3.4 it never entered the calculation. Because the configuration was described by the single quantity that can actually vary, the constraint was incorporated from the outset: motion the rod forbids is not expressible in the coordinate.

The advantage compounds with system size. A double pendulum in Cartesian coordinates requires four position variables, two rigid-length constraints, and two unknown tensions. In generalized coordinates it requires two angles.

## 4.3 The formulation is coordinate-independent

The Euler–Lagrange equation takes the same form in every coordinate system — Cartesian, polar, spherical, or arbitrary curvilinear.

Newton's second law does not. Transforming to polar coordinates introduces centrifugal and Coriolis terms that must be derived separately and included by hand; omitting one produces an error that is difficult to detect.

A domain-specific language cannot anticipate which coordinates a user will choose, so it requires a formulation indifferent to that choice. This is why `\defvar{theta}` and `\defvar{x}` are handled identically by the parser — no stage downstream needs to distinguish them.

## 4.4 Coupling is derived rather than assembled

In a double pendulum each arm's angular acceleration depends on the other's. Extracting that dependence from a force analysis is lengthy and error-prone.

From the Lagrangian it is automatic. A coupling term such as $\cos(\theta_1 - \theta_2)$ propagates correctly through the differentiation steps with no special handling.

This is why `derive_equations_of_motion` in `symbolic.py` promotes *all* coordinates to functions of time simultaneously before differentiating. Treating them sequentially would drop the cross-terms and produce equations that are plausible and wrong.

### 4.5 The structures required downstream are already present

Available directly from the Lagrangian at no additional cost:

- **Total energy**, $T + V$ — and the code recovers $T$ and $V$ from $L$ automatically (Section 6.4).
- **Conjugate momentum**, $\partial L/\partial \dot{q}$ — obtained incidentally in Step 1 of Section 3.4.
- **Conservation laws**, from the symmetries of $L$ (Section 5.5).
- **The symplectic structure** required by the integrators that prevent energy drift (Section 6.2).

A set of force vectors carries none of this. Each would require separate implementation per system.

### 4.6 The formulation generalizes

The same relation (3) applies to charged particles in electromagnetic fields, to relativistic particles, to continuous fields, and to geodesic motion in curved spacetime. This is how the `domains/` package spans sixteen fields on one engine. Newton's second law does not extend in this way.

### 4.7 Where the choice is a liability

The formulation is the weaker option when:

- **A constraint force is the quantity of interest.** If the question is whether the rod will fail, the tension is required — and recovering what the formulation eliminated requires the Lagrange multiplier machinery of Section 5.4.
- **Non-conservative forces are present.** Friction and drag dissipate energy and admit no potential, so they cannot be expressed within $T - V$ and must be reintroduced explicitly. The DSL provides `\force{}` and `\rayleigh{}` for this purpose (Section 5.6).
- **The system is trivial.** For a single particle in rectilinear motion, $F = ma$ is more direct.

For the remaining majority of problems the scalar formulation is preferable because it can be mechanized — and mechanization is the purpose of a compiler.

## 5 The symbolic engine

### 5.1 Deriving the equations

For each coordinate the engine executes the procedure of Section 3.4 symbolically:

1. Promote every coordinate to a function of time — *all coordinates simultaneously*, preserving coupling (Section 4.4).
2. Differentiate with respect to velocity, then with respect to time.
3. Subtract the derivative with respect to position.
4. Substitute back to plain symbols and simplify.

Simplification runs under a timeout, since symbolic simplification has no useful worst-case bound. On expiry the unsimplified expression is used — less readable, equally correct.

### 5.2 Solving for accelerations: the mass matrix

The Euler–Lagrange equations are always **linear in the accelerations**: each $\ddot{q}$ appears to the first power, never squared or inside a transcendental function. The system therefore admits the compact form

$$M(q)\,\ddot{q} = F(q, \dot{q}, t) \tag{13}$$

- $M$ is the **mass matrix**, an $n \times n$ array whose entry $M_{ij}$ is the coefficient of $\ddot{q}_j$ in equation $i$. Diagonal entries express each coordinate's own inertia; off-diagonal entries express inertial coupling — the extent to which accelerating one coordinate exerts an inertial effect on another. For a single particle it reduces to the mass.
- $F$ is the **generalized force vector**, comprising every term without an acceleration.

Solving requires inverting $M$. Two paths are implemented:

- **One coordinate:** direct solution. Given $A\ddot{q} + B = 0$, then $\ddot{q} = -B/A$.
- **Several coordinates:** simultaneous solution of the full linear system, with explicit matrix inversion as a fallback. Sequential solution would be invalid, since each acceleration appears in the others' equations.

If $\det M = 0$ the mass matrix is **singular** and the system has no unique solution. Physically this indicates a direction of motion carrying no inertia — a degenerate Lagrangian. The condition is detected and reported rather than propagated.

### 5.3 The Hamiltonian formulation

An alternative packaging of the same dynamics replaces velocity with momentum as the second state variable. The **conjugate momentum** is

$$p = \frac{\partial L}{\partial \dot{q}} \tag{14}$$

which for an unconstrained particle reduces to $mv$, and for the pendulum of Section 3.4 gave $ml^2\dot{\theta}$.

The change of variables is a **Legendre transform** — a systematic method for re-expressing a function in terms of its own derivative — and yields the **Hamiltonian**:

$$H = \sum_i p_i \dot{q}_i - L \tag{15}$$

with every velocity eliminated in favour of momentum. For systems whose kinetic energy is quadratic in velocity, $H = T + V$: the total energy.

The dynamics then take the symmetric first-order form of **Hamilton's equations**:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \tag{16}$$

The motivation is geometric. The space of states $(q, p)$ — **phase space**, in which a single point specifies the complete instantaneous state — carries a structure (the symplectic form) that certain integrators preserve exactly. Preserving it is what bounds long-term energy error (Section 6.2).

## 5.4 Constraints and Lagrange multipliers

A **holonomic constraint** restricts configuration alone and is written $g(q) = 0$ — a bead confined to a wire, or two masses at fixed separation.

It is imposed by augmenting the Lagrangian with the constraint and an undetermined coefficient:

$$L' = L + \sum_k \lambda_k \, g_k(q) \tag{17}$$

Each multiplier $\lambda_k$ is an additional unknown. Solving for it yields more than the constraint: $\lambda$ is proportional to the constraint force, which is how the tension eliminated in Section 4.2 is recovered when it is actually wanted.

One point requires care in implementation. The constraint must be differentiated *twice* with respect to time:

$$g = 0 \implies \dot{g} = 0 \implies \ddot{g} = 0 \tag{18}$$

If a separation is constant, its rate of change vanishes, as does the rate of change of that rate. Only the second derivative contains accelerations — the unknowns being solved for. Stopping at the first leaves the multipliers undetermined, and they propagate into the accelerations as unbound symbols. Accelerations and multipliers are therefore solved as a single combined linear system.

A **non-holonomic constraint** restricts velocities in a form that cannot be integrated to a configuration constraint:

$$\sum_j a_{ij}(q) \, \dot{q}_j + b_i(q, t) = 0 \tag{19}$$

Rolling without slipping is the standard case: the constraint restricts instantaneous velocity while leaving every position on the plane reachable. These are handled by the d'Alembert–Lagrange and Appell formulations.

The **constraint Jacobian** $\partial g / \partial q$ is rank-checked. Rank below the number of constraints indicates redundancy — the same restriction expressed more than once — which makes the augmented system unsolvable.

## 5.5 Noether's theorem

*Every continuous symmetry of the action corresponds to a conserved quantity.*

| Symmetry | Conserved quantity |
|---|---|
| Time translation ($L$ has no explicit $t$) | Energy |
| Spatial translation | Linear momentum |
| Rotation | Angular momentum |
| Galilean boost | Centre-of-mass motion |

Detection is direct. If a coordinate $q$ is absent from $L$ while its velocity $\dot{q}$ is present, then $\partial L / \partial q = 0$ and the Euler–Lagrange equation reduces to

$$\frac{\mathrm{d}}{\mathrm{d}t} \left( \frac{\partial L}{\partial \dot{q}} \right) = 0 \tag{20}$$

so the conjugate momentum is constant. Such a coordinate is termed **cyclic**, and the library identifies conservation laws by scanning for them.

For a particle in a central potential the orbital angle is cyclic — the potential is rotationally symmetric — so angular momentum is conserved. This is Kepler's equal-areas law, obtained by inspecting which symbols occur in an expression.

## 5.6 Dissipation and applied forces

Dissipative forces admit no potential and are introduced as additional terms:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} + \frac{\partial \mathcal{F}}{\partial \dot{q}_i} = Q_i \tag{21}$$

Here $\mathcal{F}$ is the **Rayleigh dissipation function**, typically $\frac{1}{2}b\dot{q}^2$ for viscous damping. The associated generalized force is $-\partial\mathcal{F}/\partial\dot{q}_i$, whose sign guarantees opposition to the motion. The term $Q_i$ denotes externally applied generalized forces — actuators, applied torques.

Implemented friction models:

| **Model** | **Form** | **Regime** |
|---|---|---|
| Viscous | $F = -bv$ | Resistance proportional to speed — fluids |
| Coulomb | $F = -\mu N\,\mathrm{sgn}(v)$ | Constant magnitude opposing motion — dry sliding |
| Stribeck | Composite | Static, Coulomb and viscous combined; reproduces stick–slip |

## 5.7 Small oscillations and normal modes

Expanded to second order about a stable equilibrium, any system reduces to

$$M\ddot{q} + Kq = 0 \tag{22}$$

where $M$ is the mass matrix of Section 5.2 and $K$ is the **stiffness matrix**, $K_{ij} = \partial^2 V/\partial q_i \partial q_j$ — the curvature of the potential at the equilibrium.

The **normal modes** are the configurations in which the entire system oscillates at a single frequency with fixed relative amplitudes. They are obtained from the **generalized eigenvalue problem**

$$K\phi = \omega^2 M\phi \tag{23}$$

whose eigenvalues give the natural frequencies $\omega$ and whose eigenvectors $\phi$ give the mode shapes.

Two pendulums coupled by a spring admit two modes: in-phase and out-of-phase oscillation. Any motion of the system is a superposition of these. The reduction of arbitrary coupled motion to a sum of independent oscillations is the reason normal-mode analysis is central to structural and molecular dynamics.

## 5.8 Stability and chaos

Equilibria are located by setting all derivatives to zero. The system is then linearized about each, and the eigenvalues of the resulting Jacobian classify it:

| **Eigenvalues** | **Classification** |
|---|---|
| All real parts negative | **Stable** — perturbations decay |
| All purely imaginary | **Marginally stable** — perturbations persist without growth |
| Any positive real part | **Unstable** — perturbations grow |

For chaotic systems a sharper measure applies. The **largest Lyapunov exponent** $\lambda$ characterizes the divergence of initially neighbouring trajectories:

$$\delta(t) \approx \delta_0 \, e^{\lambda t} \tag{24}$$

A positive $\lambda$ implies exponential sensitivity to initial conditions and a prediction horizon that degrades logarithmically with measurement precision. This is the operational definition of chaos, and it is why a double pendulum released twice from nominally identical states diverges within seconds. The exponent is estimated from trajectory data by Wolf's algorithm with periodic renormalization.

### 5.9 Singularity detection

Derived equations are analyzed before integration for conditions that will fail numerically:

- **Division by zero** — denominators possessing real roots.
- **Trigonometric singularities** — tan at $\pi/2$, cot at 0.
- **Singular mass matrix** — configurations at which $\det M = 0$ (Section 5.2).
- **Gimbal lock** — in three-angle orientation representations, configurations where two rotation axes become parallel and a degree of freedom is lost. The signature is a $1/\cos\theta$ factor. The remedy is a quaternion representation (Section 8.2).
- **Dependent constraints** — a rank-deficient constraint Jacobian.
- **Unstable equilibria** — stationary points at a maximum rather than a minimum of the potential.

Each warning reports the condition, the coordinates involved, and a suggested remedy.

### 5.10 Dimensional analysis

Units are represented as exponent vectors over base dimensions together with a scale factor. Multiplication adds exponents; division subtracts them.

This detects dimensionally inconsistent expressions — a length added to a time — at specification time rather than allowing them to reach the solver, where they would produce a numerically valid and physically meaningless result.

## 6 Numerical integration

An equation of motion gives the instantaneous acceleration. Obtaining a trajectory requires advancing the state in discrete steps, typically many thousands of them. The choice of stepping scheme materially affects the result.

### 6.1 General-purpose integration

The state vector interleaves positions and velocities as $[q_1, \dot{q}_1, q_2, \dot{q}_2, \dots]$, converting $n$ second-order equations into $2n$ first-order equations suitable for standard solvers.

Available methods: **RK45** (adaptive Runge–Kutta, the default), **RK23**, **DOP853** (eighth order), **Radau** and **BDF** (implicit), and **LSODA** (automatic switching).

**Stiffness** warrants explanation. A system is stiff when it contains dynamics on widely separated timescales — a fast oscillation superimposed on a slow evolution. Explicit methods must resolve the fastest timescale for stability reasons, even during intervals when it is dynamically irrelevant, making them prohibitively slow. Implicit methods remain stable at larger steps. The library performs a brief trial integration to detect stiffness and selects a method based on system size and integration span.

## 6.2 Symplectic integrators

Conventional methods exhibit **energy drift**. Per-step truncation errors accumulate with a consistent bias, and over long integrations a bound orbit spirals inward or outward for reasons that are entirely numerical.

**Symplectic integrators** eliminate this. Rather than minimizing local truncation error, they exactly preserve the geometric structure of phase space (Section 5.3). The consequence is that energy error oscillates within a bounded envelope indefinitely rather than accumulating — the property required for long-duration celestial mechanics.

They operate on separable Hamiltonians $H(q, p) = T(p) + V(q)$ by composing two exactly integrable sub-flows:

- a **kick**, advancing momentum under the potential, $p \to p - h\nabla V(q)$;
- a **drift**, advancing position under the kinetic term, $q \to q + h\nabla T(p)$.

Higher order is achieved by composing these with coefficients chosen so that low-order error terms cancel.

| **Method** | **Order** | **Characteristics** |
|---|---|---|
| Störmer–Verlet / Leapfrog | 2 | Explicit, time-reversible; the standard choice |
| Ruth-3 | 3 | Minimal stage count |
| Yoshida-4 | 4 | Triple-jump composition, $c_1 = 1/(2 - 2^{1/3})$, $c_2 = -2^{1/3}/(2 - 2^{1/3})$ |
| McLachlan-4 | 4 | Five stages, reduced error constant |
| PEFRL | 4 | Optimized coefficients; best long-term energy behaviour in this class |
| Yoshida-6, Yoshida-8 | 6, 8 | High-precision celestial mechanics |

Several compositions carry **negative coefficients**, meaning the integrator advances backward in time during intermediate stages. This is deliberate: the retrograde stage is what cancels the leading error terms.

An **energy drift monitor** records initial and instantaneous energy throughout a run, reporting maximum relative error and peak-to-peak oscillation, so that conservation can be verified rather than assumed.

## 6.3 Variational integrators

A structurally different approach: rather than discretizing the equations of motion, discretize the variational principle of Section 3.2 directly. The action integral becomes a sum over path segments,

$$S_d = \sum_k L_d(q_k, q_{k+1}, h) \tag{25}$$

and requiring it to be stationary yields the **discrete Euler–Lagrange equation**

$$D_2 L_d(q_{k-1}, q_k) + D_1 L_d(q_k, q_{k+1}) = 0 \tag{26}$$

solved for $q_{k+1}$ at each step by Newton iteration.

Because the scheme derives from a genuine variational principle, it inherits a discrete Noether theorem: momentum maps are conserved *exactly*, to machine precision, rather than approximately.

| **Discretization of $L_d$** | **Order** |
|---|---|
| Midpoint rule | 2 |
| Trapezoidal rule | 2 |
| Galerkin, $s$ Gauss–Legendre points | $2s$ |

The per-step nonlinear solve makes these more expensive than explicit methods, in exchange for exact structural conservation.

### 6.4 Generated code

The C++, Rust, Julia, Fortran, CUDA, MATLAB, JavaScript, WebAssembly, Arduino and ARM backends emit a fixed-step **fourth-order Runge–Kutta** integrator:

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{27}$$

with $k_1$ evaluated at the interval start, $k_2$ and $k_3$ at the midpoint, and $k_4$ at the end. It is fourth-order accurate and entirely self-contained — the relevant property for a target with no runtime library. The Python backend calls SciPy instead, since it is available.

Generated code also emits an energy function. Separating $T$ and $V$ from $L$ uses **Euler's theorem on homogeneous functions**: kinetic energy is quadratic in the velocities, so it satisfies $\sum \dot{q}\, \partial L/\partial \dot{q} = 2T$, which identifies it uniquely; $V$ is the remainder.

### 6.5 Inverse problems

Forward simulation maps parameters to trajectories. The inverse problem maps observed trajectories back to parameters — recovering a physical pendulum's length from recorded motion, for instance.

- **Parameter estimation** minimizes the residual between simulation and observation, using local optimization (fast, requires a reasonable initial guess) or differential evolution (global, no guess required). Returns covariance and confidence intervals rather than point estimates alone.
- **Sensitivity analysis** computes **Sobol indices**, which decompose output variance among the inputs. First-order indices give each parameter's independent contribution; total-order indices additionally capture its interactions.
- **Uncertainty quantification** uses MCMC to sample the posterior distribution over parameters, yielding credible intervals, and bootstrap resampling as an independent check.

## 7 Energy as a correctness criterion

Energy conservation is the primary validation check available, because a simulation that dissipates energy numerically produces trajectories that appear plausible and are wrong.

For standard systems the library uses closed-form expressions.

**Simple pendulum.**

$$T = \frac{1}{2}ml^2\dot{\theta}^2, \qquad V = -mgl\cos\theta \tag{28}$$

**Double pendulum.**

$$T = \frac{1}{2}m_1 l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\left(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1 l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) \tag{29}$$

$$V = -m_1 g l_1\cos\theta_1 - m_2 g\,(l_1\cos\theta_1 + l_2\cos\theta_2) \tag{30}$$

**Spherical pendulum.**

$$T = \tfrac{1}{2}ml^2\left(\dot{\theta}^2 + \sin^2\theta\,\dot{\phi}^2\right), \qquad V = mgl(1-\cos\theta) \tag{31}$$

**Harmonic oscillator.**

$$T = \tfrac{1}{2}mv^2, \qquad V = \tfrac{1}{2}kx^2 \tag{32}$$

The double pendulum's cross term $2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)$ is the inertial coupling between the arms, and it is the source of the system's chaotic behaviour. It follows immediately from writing down the kinetic energy; deriving it from a force analysis requires substantially more work.

Potential energies are offset so that the minimum reads zero — a pendulum at rest has $V = 0$. This affects only the plotted baseline, since potential energy is defined up to an additive constant and only differences are physically meaningful.

# 8 Domain reference

The `domains/` package implements sixteen fields beyond the core engine. Each entry states its scope in one sentence; the material following documents coverage.

## 8.1 Kinematics

**Motion under constant acceleration, treated without reference to its causes.**

Five equations describe all such motion:

$$v = v_0 + at \tag{33}$$
$$x = x_0 + v_0t + \tfrac{1}{2}at^2 \tag{34}$$
$$v^2 = v_0^2 + 2a(x-x_0) \tag{35}$$
$$x = x_0 + \tfrac{1}{2}(v+v_0)t \tag{36}$$
$$x = x_0 + vt - \tfrac{1}{2}at^2 \tag{37}$$

Each omits exactly one of the six quantities $\{x_0, x, v_0, v, a, t\}$, so any three knowns determine the rest. The solver selects the appropriate equation and reports the derivation steps — the module is written for instructional use.

Also implemented: one-dimensional motion (uniform, uniformly accelerated, free fall, vertical throw); two-dimensional motion by independent component decomposition; projectile motion with range, apex, flight time and impact velocity; and relative motion via $v_{AC} = v_{AB} + v_{BC}$.

## 8.2 Rigid body dynamics

**Bodies that rotate without deforming — gyroscopes, spinning tops, spacecraft.**

Orientation is represented in one of two ways:

- **Euler angles** $(\phi, \theta, \psi)$: precession, nutation and spin, in the ZYZ convention. Geometrically interpretable, but subject to gimbal lock at $\theta = 0, \pi$, where two axes align and the parameterization degenerates.
- **Quaternions** $(q_0, q_1, q_2, q_3)$: four parameters subject to the unit-norm constraint $q_0^2+q_1^2+q_2^2+q_3^2 = 1$. Singularity-free at every orientation, at the cost of one redundant parameter and less direct interpretation.

The symmetric top has Lagrangian

$$L = \tfrac{1}{2}I_1\left(\dot\theta^2 + \dot\phi^2 \sin^2\theta\right) + \tfrac{1}{2}I_3\left(\dot\psi + \dot\phi\cos\theta\right)^2 - Mgl\cos\theta \tag{38}$$

Rotational kinetic energy is $\frac{1}{2}\omega^{\mathsf{T}} I\omega$, where $I$ is the inertia tensor — the rotational analogue of mass, encoding the distribution of mass about the rotation axes. Angular velocity from quaternion rates is $\omega = 2E\dot q$.

## 8.3 Central forces and orbital mechanics

**Motion under a force depending only on separation — gravitation and electrostatics.**

For $V(r)$ the Lagrangian is $\frac{1}{2}m(\dot r^2 + r^2\dot\phi^2) - V(r)$. The angle $\phi$ is cyclic, so angular momentum $L = mr^2\dot\phi$ is conserved (Section 5.5), and the two-dimensional problem reduces to one-dimensional radial motion in the **effective potential**

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2} \tag{39}$$

The second term is the **centrifugal barrier**, representing the angular momentum that must be overcome to approach the centre.

Turning points satisfy $E = V_{\text{eff}}(r)$, and their number and arrangement classify the orbit as circular, elliptical, parabolic or hyperbolic. Includes the Kepler problem and apsidal precession rates for non-inverse-square potentials.

## 8.4 Collisions and scattering

**Impulsive interactions and deflection by a potential.**

Momentum is conserved in all collisions; kinetic energy only in elastic ones. The **coefficient of restitution** $e$ parameterizes the range: $e = 1$ elastic, $0 < e < 1$ inelastic, $e = 0$ perfectly inelastic with the bodies coalescing. Includes centre-of-mass frame transformations and impulse calculations.

Scattering treats a particle incident from infinity, deflected through angle $\theta$, departing to infinity. Covers the impact-parameter-to-angle relation, differential and total cross-sections, and Rutherford scattering.

## 8.5 Variable-mass systems

**Systems whose mass changes during motion — principally rockets.**

Newton's second law in the form $F = ma$ does not apply. The correct statement retains the momentum flux term:

$$F_{\text{ext}} = m\frac{\mathrm{d}v}{\mathrm{d}t} + v_{\text{rel}}\frac{\mathrm{d}m}{\mathrm{d}t} \tag{40}$$

The second term accounts for momentum carried by mass entering or leaving the system. A rocket accelerates by expelling mass, not by reaction against an external medium. Includes the Tsiolkovsky equation, ejection and accretion, conveyor belts and chain dynamics.

## 8.6 Continuum mechanics and classical fields

**Systems with continuously distributed degrees of freedom — strings, membranes, fields.**

The configuration is a field $\phi(x, t)$ rather than finitely many coordinates, and the action becomes a double integral over space and time. The Euler–Lagrange equation generalizes to

$$\frac{\partial\mathcal{L}}{\partial\phi} - \frac{\partial}{\partial t}\frac{\partial\mathcal{L}}{\partial\phi_t} - \frac{\partial}{\partial x}\frac{\partial\mathcal{L}}{\partial\phi_x} = 0 \tag{41}$$

yielding the wave equation. Covers vibrating strings, membrane vibration and the stress–energy tensor.

## 8.7 Fluid dynamics

**Free-surface and splashing flows, by particle methods.**

**Smoothed Particle Hydrodynamics** discretizes a fluid into particles whose properties are distributed over a finite smoothing radius $h$ rather than concentrated at points. Field values at any location are recovered by weighted summation over neighbours within that radius, which bounds the cost per particle. The method handles free surfaces and topological changes naturally, where grid methods require special treatment.

Three kernels are used:

| **Kernel** | **Form** | **Application** |
|---|---|---|
| Poly6 | $\frac{315}{64\pi h^9}\left(h^2 - r^2\right)^3$ | Density |
| Spiky gradient | $-\frac{45}{\pi h^6}(h - r)^2\,\hat{r}$ | Pressure force |
| Viscosity Laplacian | $\frac{45}{\pi h^6}(h - r)$ | Viscous force |

Density is $\rho_i = \sum_j m_j W(r_{ij}, h)$, with pressure from a Tait-form equation of state $p = k(\rho - \rho_0)$. Forces comprise pressure (pairwise symmetrized), viscosity and gravity; integration is semi-implicit Euler. Boundary conditions include no-slip and free-slip walls, periodic and open.

## 8.8 Electromagnetism

**Charged particle dynamics, wave propagation and plasma behaviour.**

The charged-particle Lagrangian incorporates the scalar and vector potentials,

$$L = \tfrac{1}{2}mv^2 - q\varphi + q\,\vec{v}\cdot\vec{A} \tag{42}$$

reproducing the **Lorentz force** $\vec{F} = q(\vec{E} + \vec{v}\times\vec{B})$.

The conjugate momentum is $\vec{p} = m\vec{v} + q\vec{A}$, not $m\vec{v}$ — a direct illustration that momentum in this formalism is defined by $\partial L/\partial\dot{q}$ and need not coincide with the elementary definition.

Also implemented: cyclotron frequency and Larmor radius; magnetic mirrors with loss cones; Penning traps (cyclotron, axial, magnetron and modified-cyclotron frequencies, and the invariance theorem); guiding-centre drifts (grad-$B$, curvature, polarization, $\vec{E}\times\vec{B}$); plasma frequency and Debye length; electromagnetic waves (impedance, intensity, energy density, radiation pressure); antennas (Hertzian and half-wave dipoles, gain, effective aperture); rectangular waveguides (cutoff frequencies, guide wavelength, phase and group velocity, TE/TM impedance); skin effect; and the Biot–Savart law.

## 8.9 Special relativity

**Dynamics at speeds comparable to $c$.**

The relativistic Lagrangian $L = -mc^2\sqrt{1 - v^2/c^2} - V$ is equivalent to extremizing proper time along the worldline: of all paths between two events, the physical one maximizes the elapsed time measured by the traveller.

The **Lorentz factor** $\gamma = 1/\sqrt{1 - v^2/c^2}$ governs all relativistic corrections, approaching 1 at low speed and diverging as $v \to c$. Core relations: $p = \gamma mv$, $E = \gamma mc^2$, $T = (\gamma - 1)mc^2$, and the invariant $E^2 = (pc)^2 + (mc^2)^2$.

Four-vectors are classified as timelike, spacelike or lightlike by the sign of their invariant. Also covers Lorentz boosts, relativistic velocity addition, time dilation, length contraction, rapidity, relativistic collisions and threshold energies, Mandelstam $s$, longitudinal and transverse Doppler shift, aberration, Compton scattering, synchrotron radiation, Thomas precession, and the twin paradox for both constant velocity and constant proper acceleration.

## 8.10 General relativity

**Gravitation as spacetime curvature; black holes and cosmology.**

**Schwarzschild geometry** (static, spherically symmetric):

$$\mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right) c^2 \mathrm{d}t^2 + \left(1 - \frac{r_s}{r}\right)^{-1} \mathrm{d}r^2 + r^2\, \mathrm{d}\Omega^2 \tag{43}$$

with $r_s = 2GM/c^2$. Provides gravitational redshift, the photon sphere at $1.5\, r_s$, the innermost stable circular orbit at $3\, r_s$, tidal acceleration, surface gravity and Hawking temperature. Geodesics — the trajectories of free particles, which are the straightest available paths in curved geometry — are integrated from an effective potential, with separate treatment for massive and massless particles.

**Kerr geometry** (rotating): inner and outer horizons, the ergosphere (within which frame dragging makes static observers impossible), frame-dragging rate, and spin-dependent prograde and retrograde ISCO radii.

**Gravitational lensing**: deflection $\alpha = 4GM/(c^2 b)$, Einstein radius, magnification and time delay.

**FLRW cosmology**: Hubble parameter as a function of redshift, Hubble time and distance, comoving, luminosity and angular-diameter distances, lookback time, cosmic age, critical density and the deceleration parameter.

## 8.11 Quantum mechanics

**Semiclassical methods — the regime where classical quantities determine quantum results.**

The **WKB approximation** treats the wavefunction as a slowly modulated wave with local wavenumber set by the classical momentum $p(x) = \sqrt{2m(E - V(x))}$. The **Bohr–Sommerfeld condition** then quantizes the classical action:

$$\oint p\, \mathrm{d}x = \left(n + \tfrac{1}{2}\right) h \tag{44}$$

Only orbits whose enclosed action is a half-integer multiple of Planck's constant are permitted, which is the origin of discrete energy levels. This is the point at which the classical machinery of the rest of the library yields quantum results.

Exactly solved systems: the harmonic oscillator ($E_n = \hbar\omega(n + \frac{1}{2})$, Hermite wavefunctions, uncertainty products); infinite and finite square wells; step potentials; delta-function barriers; and the hydrogen atom ($E_n = -13.6\ \mathrm{eV}/n^2$, Bohr radii, degeneracy, transition wavelengths, spectral series).

**Tunneling** — transmission through a barrier exceeding the particle's energy, classically forbidden — covers rectangular barriers, WKB transmission, the Gamow factor, alpha decay rates, double-well splitting and resonant tunneling. **Ehrenfest's theorem** propagates expectation values classically, establishing the correspondence limit.

## 8.12 Thermodynamics

**Heat engines, equations of state and phase behaviour.**

**Carnot efficiency** $\eta = 1 - T_c/T_h$ bounds every heat engine operating between two reservoirs — a consequence of the second law rather than an engineering limitation, and the reason thermal power generation necessarily rejects heat.

Also implemented: the Otto cycle ($\eta = 1 - r^{1-\gamma}$, monotonic in compression ratio), Diesel and Rankine cycles; the van der Waals equation $\left(P + an^2/V^2\right)(V - nb) = nRT$ with critical point and compressibility factor; Redlich–Kwong; phase transitions via Clausius–Clapeyron; Debye and Einstein heat capacity models; and the four Maxwell relations.

## 8.13 Statistical mechanics

**Derivation of macroscopic thermodynamics from microscopic states.**

The **Boltzmann distribution** assigns probability proportional to $e^{-E/k_B T}$, with the **partition function** $Z = \sum_i e^{-E_i/k_B T}$ as the generating quantity from which thermodynamic properties follow.

Includes the Maxwell speed distribution with most-probable, mean and RMS speeds; ideal gas pressure, internal energy, heat capacities, Sackur–Tetrode entropy and free energies; quantum oscillator ensembles; the Ising model with Metropolis Monte Carlo and the exact two-dimensional critical temperature; and Fermi–Dirac and Bose–Einstein occupation statistics with Fermi energy and BEC critical temperature.

## 8.14 Solid mechanics

**Deformation and failure of engineering materials.**

- **Elasticity** — Hooke's law for isotropic, transversely isotropic, orthotropic, monoclinic and triclinic symmetry classes.
- **Stress and strain** — principal stresses, invariants $I_1, I_2, I_3, J_2, J_3$, von Mises and Tresca yield criteria, Mohr's circles in two and three dimensions, tensor transformations, stress concentration factors and strain rosette reduction.
- **Beams** — Euler–Bernoulli theory for slender beams and Timoshenko theory where shear deformation is significant; deflection, slope, bending moment and shear; composite, tapered and curved beams.
- **Plates** — Kirchhoff–Love for thin plates and Mindlin–Reissner for thick; rectangular and circular solutions, buckling and natural frequencies.
- **Fracture** — stress intensity factors for Modes I, II and III; energy release rate; the $J$-integral; CTOD; the Griffith criterion (propagation when released strain energy exceeds the surface energy cost); Paris law for fatigue crack growth; and plastic zone size.
- **Fatigue** — failure under cyclic loading below the static strength. S–N curves $\sigma_a = \sigma_f'(2N_f)^b$, strain-life, mean stress correction and cumulative damage.
- **Viscoelasticity** — Maxwell, Kelvin–Voigt and Standard Linear Solid models; generalized Maxwell via Prony series; creep and relaxation functions; complex modulus; time–temperature superposition; and Boltzmann superposition.
- **Thermal stress** — $\varepsilon = \alpha\Delta T$ linear and $3\alpha\Delta T$ volumetric, with thermoelastic coupling. Constrained thermal expansion generates stress independent of applied load.

## 8.15 Optics

**Propagation, imaging and interference of light.**

Geometric optics (ray tracing, Snell's law, reflection, thin and thick lenses), wave optics (interference and diffraction), polarization via Jones matrices and Stokes parameters, and fibre optics.

## 8.16 Acoustics

**Generation, propagation and confinement of sound.**

Wave equation solutions in one, two and three dimensions; resonance and standing waves; acoustic impedance and boundary reflection; the Doppler effect; room acoustics and Sabine reverberation; and ultrasonics.

# 9 Summary

The library rests on a single organizing principle:

> *A mechanical system is specified by one scalar function — kinetic energy minus potential energy — and everything else follows by differentiation.*

Given $L = T - V$, a fixed procedure yields the equations of motion; inverting the mass matrix yields accelerations; a Legendre transform yields the energy; absent coordinates yield conservation laws; structure-preserving integrators yield trajectories that respect those conservation laws over arbitrarily long runs; and the same symbolic expressions compile to thirteen target languages.

The Newtonian formulation yields identical predictions. It requires, however, that a person derive them — once per system, once per coordinate choice, by hand, correctly.

That is precisely the step a compiler cannot perform on the user's behalf, and it is why the library is built on the other formulation.